may want to have a copy of a TTL data book, such as Reference 1.2. You need not be familiar with the 4116 RAM or the 6502 microprocessor. Special sections in Chapters 5 and 6 will cover these devices. However, you should have a basic knowledge of microprocessor and microcomputer architecture. And of course you should be familiar with the binary and hexadecimal number systems.

Concerning your Apple II background, you should be familiar with the *Apple II Reference Manual* (Reference 1.1).

REVISIONS

There have been several circuit revisions to the Apple II since its introduction in 1977. The revisions are discussed in detail in Appendix B. Here in Chapter 1 we simply summarize the changes and establish a revision nomenclature for use throughout the book.

There are two categories of the Apple II mother board: *Non-RFI* (the early mother boards) and *RFI* (recent mother boards that have been designed to reduce radio-frequency interference).

Non-RFI mother boards have the part number 820-0001-XX where XX is the revision. The first non-RFI mother board was Revision 0; we will call it simply *Rev. 0*. It had only four HIRES colors. Revision 0 also lacked color killer and power-on reset circuits.

Revision 1 came next and added two more HIRES colors (for the current total of six). Revision 1 also added color killer and power-on reset circuits and made other small changes. Revisions 2, 3, and 4 had the same circuit as Rev. 1. In this book, we include them with Rev. 1.

The next significant change occurred at revision 7—the memory jumper blocks were removed and the character generator IC was changed.

The next significant change occurred with the switch to the RFI mother board. This board has the part number 820-0044-XX where XX is the revision. All revisions of this board to date (through Rev. D) have the same functional circuit. We refer to it simply as *RFI*.

The part numbers are found either along the left edge of the mother board or under the 6502 IC. See Appendix B for more details. In summary, the mapping from mother board part numbers to our nomenclature is shown in Table 1-1.

Table 1-1. Mapping From Mother Board Part Numbers

	Part No.	Revision
Non-RFI	820-0001-00	Rev. 0
	820-0001-01 820-0001-02 820-0001-03 820-0001-04	Rev. 1
	820-0001-07 & up	Rev. 7
RFI	820-0044-01 820-0044-C 820-0044-D	RFI

IC AND SIGNAL NOMENCLATURE

The ICs on the mother board are designated by their location on an X-Y grid. The grid coordinates consist of the letters A through K along the left edge of the board

APPENDIX B

Apple's Revisions

Like many electronic products that are manufactured for several years, the Apple II has undergone revision from time to time. Some revisions are product enhancements, such as adding more graphics colors. Some changes, like adjusting the cassette input gain, improve performance or reliability. Other revisions reduce cost by taking advantage of changing technology. An example is elimination of the ability to use the old 4K RAMs. One revision expands the market by providing optional compatibility with European television. Finally, some changes are mandated by government regulation. Here we speak of the redesign required for compliance with FCC rules on radio frequency interference.

The list of revisions may seem long, but the Apple II has actually sustained little functional change since its initial release in 1977. This can be credited to the foresight and innovation of the original design.

In this appendix we describe mother board and keyboard circuit changes that have occurred since the Apple II's initial design. We also note which changes occurred at which revision level and how to determine the revision of a particular Apple.

The circuit description in the body of this book covers the current Revision D RFI mother board and the two-piece keyboard. When earlier revisions require different descriptions, those descriptions are provided in this appendix.

OVERVIEW OF REVISIONS

Mother Board

The Apple II mother board has been manufactured in two basic varieties: *Non-RFI* and *RFI*. The non-RFI board was produced until 1981 when Apple switched to the RFI board. Radio frequency interference (RFI) refers to the electromagnetic noise given off by high-speed circuits such as computers. This noise often interferes with radio and tv sets. The FCC can restrict the sale of computers that exceed certain RFI limits. For this reason, Apple redesigned the mother board to reduce RFI. This redesigned board is referred to as the "RFI Mother Board." Earlier boards are referred to as "Non-RFI Mother Boards."

The RFI redesign involved such things as interchanging the location of + 5 volt and ground buses, isolating some circuit grounds, and adding ferrite beads. There were few functional changes. (Note that the Apple II RFI redesign involved more than the mother board. Part of the change involved shielding the plastic case and using shielded cable to the peripherals.)

Within the two major categories of RFI and non-RFI, there are additional revisions. We will list these in later sections.

Non-RFI—The non-RFI mother boards are designated by the nine digit part number 820-0001-XX. The "XX" is an incrementing revision number that starts with 00. The part number is found near the F1 location (left edge of board) on later revisions, and under the 6502 microprocessor IC on earlier revisions. (In this latter case, it is necessary to remove the 6502 to see the part number.) On the first or "Rev. 0" board, there may be no part number at all.

It is possible to determine the revision of a non-RFI mother board by examining certain features on the board. This is handy if the part number is not visible. A Rev. 0 board does not have an Auxiliary Video Pin. This pin is a single vertical pin located between the four-pin Auxiliary Video Connector and the video level adjust potentiometer (see Fig. B-1).

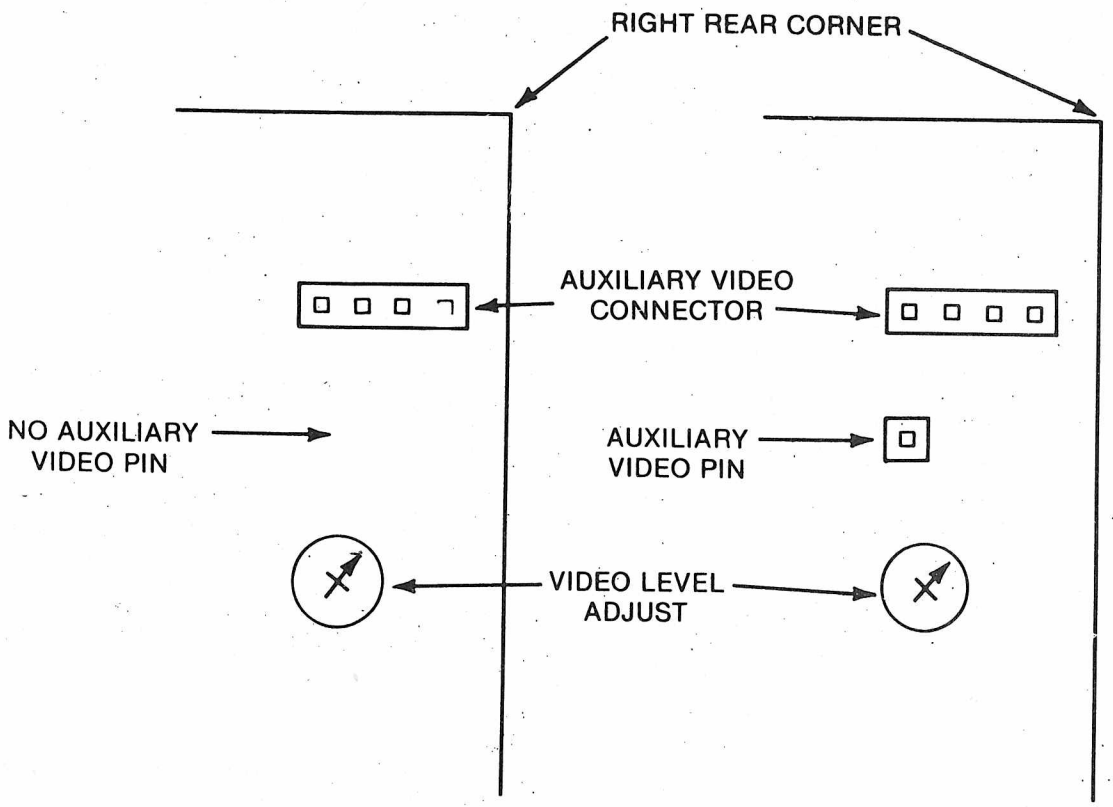


Fig. B-1. Identification of early non-RFI mother boards.

Non-RFI revisions 0 through 6 all have Memory Select Jumpers at locations D1, E1, and F1. Revision 7 does not have these jumpers.

There are no significant circuit differences between the revision 1 through 6 mother boards. Also, we have not found any distinguishing visible features (other than the part number) that can be used to differentiate these revisions. *In this book,*

we refer to the revision 1 through 6 non-RFI mother boards as simply "Rev. 1." Similarly, *revision 7 and higher non-RFI boards are referred to as "Rev. 7."*

RFI—The RFI mother boards are identified by the part number 820-0044-XX. The "XX" is the revision and to date (July 1982) the latest revision is "D." There have been previous revisions of "01" and "C." The part number is found near the F1 location (left edge of board). The differences between RFI revisions involve locations and quantities of ferrite beads and other changes intended to further suppress RFI emissions. To date there have been no functional circuit changes between RFI revisions. *We refer to all RFI mother boards (through revision D) as simply "RFI."*

Keyboard

Single-piece—The early Apple II computers contain a single-piece keyboard; the keys and electronic components are mounted on a common PC board. This keyboard uses the National Semiconductor MM5740-AAE encoder IC which is no longer manufactured. The single-piece keyboard is described in this appendix.

Two-piece—At about the time that Apple introduced the Rev. 7 non-RFI mother board, they also introduced a two-piece keyboard. In this design, the keys are mounted on one PC board, and the electronic components are mounted on another. The two boards then plug together. There have been revisions of the board containing the keys. These revisions allow for key-switches from different manufacturers. These different boards, however, are all plug-compatible with the electronics board. The two-piece keyboard uses the General Instrument AY-5-3600 encoder IC and is described in Chapter 7.

Apple II Plus

The Apple II Plus is not a revision in the same sense as the mother board circuit revisions listed in this appendix. The difference between the Apple II and the Apple II Plus is the choice of firmware plugged into the mother board. Integer BASIC is on the Apple II mother board, and Applesoft BASIC is on the Apple II Plus mother board. In this book we have made no distinction between the Apple II and the Apple II Plus.

DETAILED CIRCUIT ANALYSIS

In this section we work backwards in time, describing circuit differences between revisions of the Apple. Remember that the RFI mother boards (part numbers 820-0044-01, -C, and -D) are described in the body of the book. It is with respect to the RFI board that we describe circuit variations.

Revision 7 Non-RFI Mother Board

The revision 7 non-RFI mother boards are identified by part numbers 820-0001-07 and up.

Color Killer—Early Rev. 7 boards do not have IC A14 (Fig. C-19*). These boards use transistor Q6 as a color killer. When TEXT MODE is high, base current through R27 drives Q6 into saturation. This shunts COLOR BURST to ground. This use of Q6 is not fully effective and some tv sets still display tinted text characters. In later Rev. 7 boards, A14-1 and A14-4 are added to force COLOR BURST to a low when TEXT MODE is high. For these boards, the last waveform of Fig. 4-3 is always low. Note that A14 is inserted into the circuit differently on Rev. 7 boards than it is on RFI boards. Also note the use of COLOR REF versus $\overline{\text{COLOR REF}}$.

Soft 5—Rev. 7 and earlier boards use high outputs of gates A2-8 and A2-11 instead of R28 to pull up unused logic inputs (Fig. C-2*).

Data Transceiver—Rev. 7 and earlier boards use two 8T28 quad transceivers on the data bus (H10 and H11 in Fig. C-8) instead of a single 8304 octal transceiver.

H2 Inverter—Rev. 7 boards use Q7 and associated resistors (Fig. C-19*) instead of A14-13 (Fig. C-20*) to invert signal H2. Q7 is located near IC B13.

Vertical Sync—Rev. 7 boards do not use A14-4 and A14-10 in the vertical sync circuit (Fig. C-19*). This results in a slightly different vertical sync pulse waveform as shown in Fig. B-2*. The waveforms that differ between Rev. 7 and RFI boards are B14-1, C13-C, SYNC, and the COMPOSITE VIDEO OUTPUT. For Rev. 7, Fig. B-2* replaces the vertical sync portion of Fig. 4-2*.

Auxiliary Video Pin—The Auxiliary Video Pin on some Rev. 7 boards is mounted in a four-pin connector as shown in Fig. C-19*. Two pins in this connector connect to unused solder pads 3 and 4 nearby. The fourth pin connects to signal AN2 and solder pad 2.

Video Filter—Rev. 7 and earlier boards do not have the high frequency filter consisting of L7 and C16 in the composite video output (Fig. C-19* and C-20*).

Miscellaneous—On Rev. 7 boards, the Clear input for text mode shift register A3 connects to SOFT 5 via bow tie 5.

Revision 1 Non-RFI Mother Board

The revision 1 non-RFI mother boards are identified by part numbers 820-0001-01 through -06.

Color Killer—Rev. 1 boards use Q6 as their only color killer. The operation of Q6 was described previously under Rev. 7.

Text Points—Rev. 1 and Rev. 0 boards do not have test points 3 and 4 or solder pad 1 (Figs. C-2*, C-12*, and C-16*).

Memory Jumper Blocks—Rev. 1 and Rev. 0 boards have 16-pin Memory Jumper Blocks at locations D1, E1, and F1 (Fig. C-4*). These blocks allow 16K-bit *or* 4K-bit RAMs to be used. The jumper blocks in Fig. C-4* are drawn in the configuration for use of *all* 16K RAMs. In this configuration, the memory system operates as described in Chapter 5. The following ICs and gates are part of the 4K/16K memory arrangement of Rev. 1 and Rev. 0 boards: H1-11, H1-3, C14-3, and E2. The IC E2 may be removed if you are using all 16K RAMs in a Rev. 1 or Rev. 0 Apple.

RAM Address 6 Termination—R31 and R32 which terminate signal line RA6 are not present on Rev. 1 or Rev. 0 boards.

Solder Pads 7 and 8—Solder pads 7 and 8 (Fig. C-11*) are not present on Rev. 1 or Rev. 0 boards. These pads are used to connect signals to peripheral I/O slot 7 for use by European color video generators.

Cassette Input—The gain of the cassette input circuit is different on Rev. 0 and some Rev. 1 boards (Fig. C-12*).

Test Connector—Rev. 1 and Rev. 0 boards do not have a multipin test connector near IC B2 (Fig. C-3*).

Character Generator—On Rev. 1 and Rev. 0 boards, IC A5 is a 2513 character generator (Fig. C-15*). A 2513 is really a ROM containing the same set of 64 ASCII characters that have been programmed into the 2316B ROM of the Rev. 7 and RFI boards. The 2513 cannot be directly replaced by a user-supplied EPROM, however, due to its supply voltages and pinouts.

The 2513 supplies only five outputs; the five horizontal dots of a character. As a result, the B and H inputs of shift register A3 are tied to ground. Bow ties 9, 10, and

11 are not present on Rev. 1 or Rev. 0 boards. Apart from these differences, the text mode operation of Rev. 1 and Rev. 0 boards is the same as described in Chapter 7.

Video Sync—Rev. 1 mother boards do not have IC A14 (compare Figs. C-18* and C-20*.) The result is a horizontal sync pulse twice as wide on Rev. 1 as on Rev. 7 and RFI boards. Fig. B-3 shows as typical the horizontal sync pulse that follows the vertical sync pulse. When dealing with a Rev. 1 board, use Fig. B-3 to replace the appropriate parts of Fig. 4-2*.

Fig. B-4 *shows the wider sync pulse and COLOR BURST in detail. When dealing with Rev. 1, Fig. B-4 replaces Fig. 4-3. In summary, the video signals that differ between Rev. 1 and RFI boards are B14-1, C13-C, C13-D, SYNC, COLOR BURST, and the COMPOSITE VIDEO OUTPUT.

Miscellaneous—The following items also differ between Rev. 1 and RFI boards: SOFT 5, Data Transceiver, and Video Filter. These items were described previously under Rev. 7.

Revision 0 Non-RFI Mother Board

The revision 0 non-RFI mother boards are identified as shown in Fig. B-1.

Power-On Reset—The Rev. 0 mother board does not have a power-on reset circuit. This function is provided by type 555 timer A13 on Rev. 1 and later boards (Fig. C-13*). This means that Rev. 0 users may have to press the reset key to initialize the 6502. This is not necessary, however, if a peripheral is installed that drives a reset pulse onto the reset line at power up. The Apple II floppy disk controller card is such an example.

Without power-on reset, the keyboard strobe flip-flop (B10-9) may be set when power is applied to a Rev. 0 board. As a result, the first keyboard command may contain an extra leading character. Note: If an autostart monitor ROM is installed in a Rev. 0 board, reset routines in the ROM will reset the keyboard strobe.

Color Killer—Rev. 0 boards use neither IC A14 nor transistor Q6 for a color killer. Without a color killer, COLOR BURST is not suppressed in text mode. This results in text characters with a color tint or fringe when using a color tv set.

Four HIRES Colors—The Rev. 0 mother board displays only four HIRES colors: black, white, violet, and green. By comparing Figs. C-14* and C-15*, you can see that some components are absent from Rev. 0. The key component not present is flip-flop A11-9. Recall from the HIRES discussion in Chapter 8 that A11-9 provides a delayed version of the HIRES serial bit stream from B4-15. Data selector A9 then selects either the delayed or nondelayed bit stream under control of DL7. When the delayed bit stream is selected, blue and orange are provided.

On Rev. 0 boards, there is no delayed bit stream, so there are no blue and orange colors. Data bit DL7 is ignored in HIRES mode on a Rev. 0 mother board.

Also on Rev. 0 boards, DL7 does not connect to A9 (this connection is via A8 and A10 on Rev. 1 and later boards). The data and select inputs of A9 are also arranged differently on Rev. 0 boards, with A12-10, A12-13, and A11-5 absent. Apart from the two missing HIRES colors, the graphics and text operation of Rev. 0 is the same as that of the RFI mother board described in Chapter 8.

24K Memory Size Problem—There is a problem on the Rev. 0 mother board that affects Apples with 20K or 24K bytes of RAM. The problem causes BASIC to "think" that there is more RAM in the computer than there is. This problem is corrected on the Rev. 1 board by gate C14-3 (Fig. C-4*). See page 72 of the *Apple II Reference Manual* for more details.

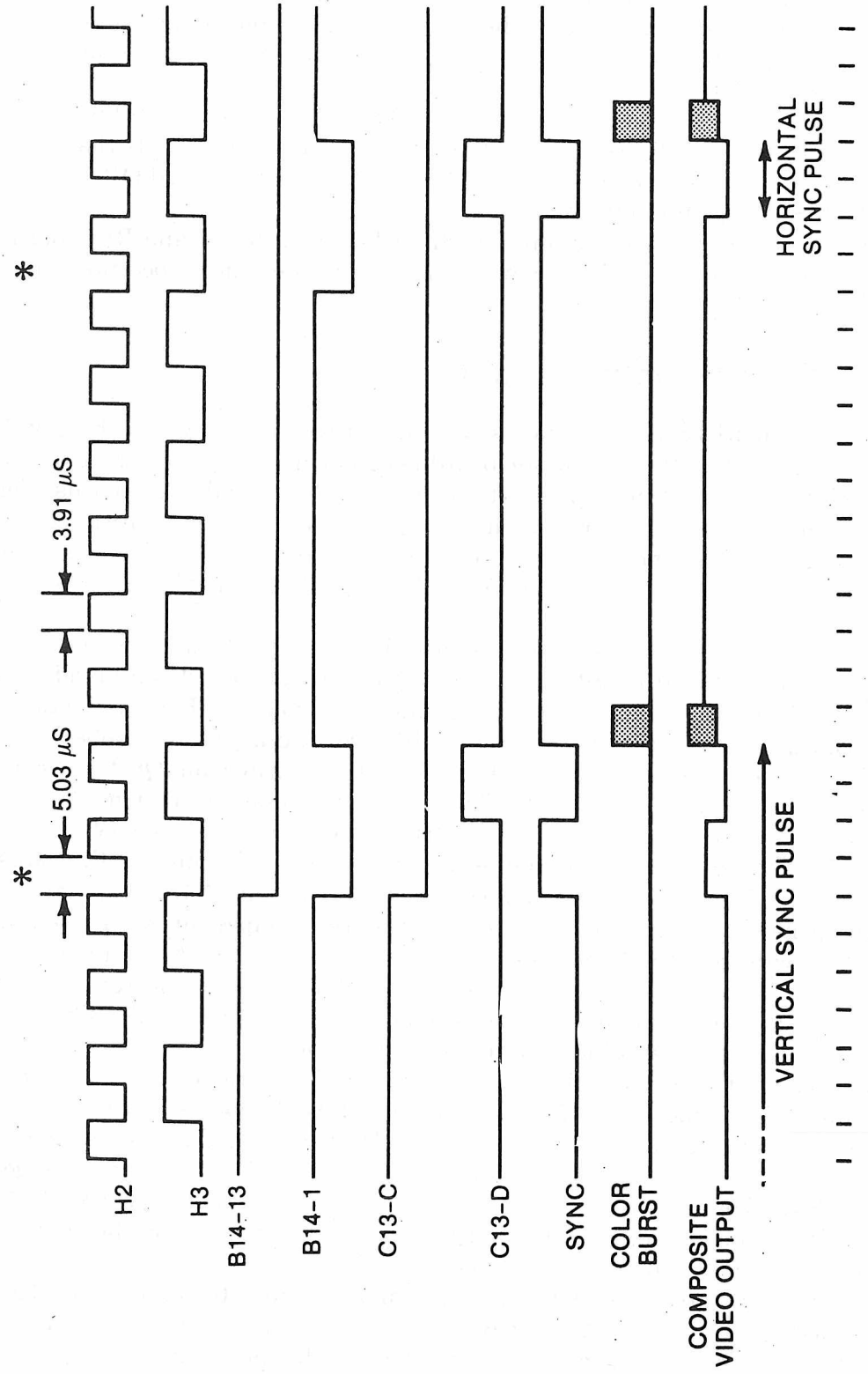


Fig. B-3. Vertical and horizontal sync—Rev. 1.

European TV Standards—Bow ties 12 and 16, and solder pads 13, 14, and 15 on Rev. 1 and later boards can be rearranged to convert the vertical frequency to 50 Hz for use with European tv sets. These pads and bow ties are not present on Rev. 0 boards. The circuit differences may be seen in Figs. C-3*, C-17*, and C-18*. The operation of the video address generator on a Rev. 0 board is the same as described in Chapters 3 and 4 for RFI boards.

Auxiliary Video Pin—Rev. 0 boards do not have an Auxiliary Video Pin (compare Figs. C-17* and C-18*).

Slot 7—Signals SYNC and COLOR REF do not connect to peripheral I/O slot 7 on Rev. 0 boards (Fig. C-11*).

Video Sync—Rev. 0 boards do not have ICs A14 and B14 (compare Figs. C-17* and C-20*). Since there is no A14, the Rev. 0 horizontal sync pulse is twice as long as the RFI horizontal sync pulse (see Figs. B-4 and B-6*). When dealing with the Rev. 0 mother board, substitute Fig. B-4 for Fig. 4-3.

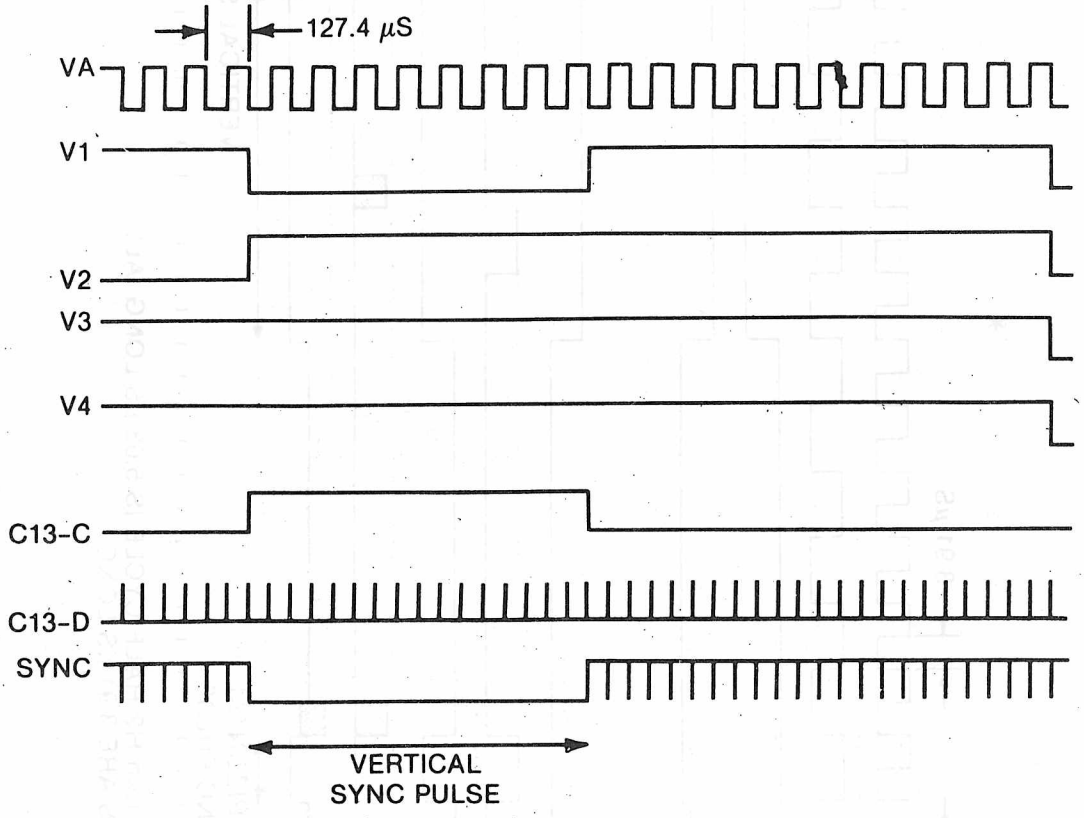


Fig. B-5. Vertical sync—Rev. 0.

Since there is no B14, the Rev. 0 vertical sync pulse has no serrations and is four times as long as the RFI vertical sync pulse (compare Figs. B-5 and 4-1*). Note that to the scale of the figures, the Rev. 0 and RFI vertical sync pulses both start at the same time relative to the video address. When dealing with the Rev. 0 mother board, substitute Figs. B-5 and B-6 for the appropriate sections of Figs. 4-1* and 4-2*.

Miscellaneous—The following items also differ between Rev. 0 and RFI boards (they were described previously under Rev. 1):

Test Points
Memory Jumper Blocks

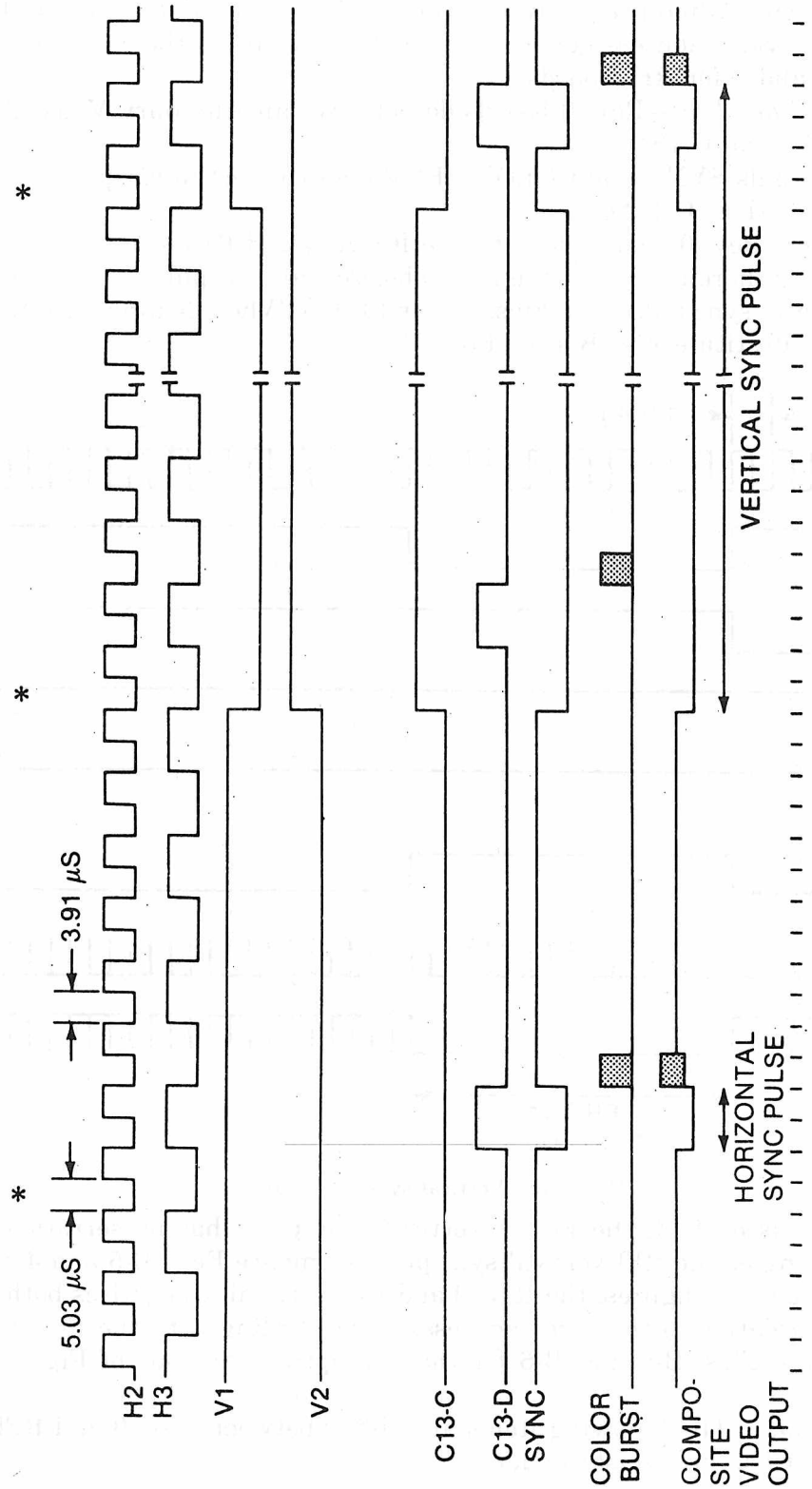


Fig. B-6. Vertical and horizontal sync—Rev. 0.

RA6 Terminator
Solder Pads 7 and 8
Cassette Input
Test Connector
Character Generator

The following additional items of difference between Rev. 0 and RFI boards were described previously under Rev. 7:

SOFT 5
Data Transceiver
Video Filter

Table B-1 lists a summary of the major differences between mother board revisions.

Table B.1 Summary of Major Mother Board Revisions

	Rev. 0	Rev. 1	Rev. 7	RFI
Color Killer	No	Q6	Q6 or A14-4	A14-1
Horizontal Sync Pulse	7.8 μS	7.8 μS	3.9 μS	3.9 μS
Vertical Sync Pulse	1.02 mS	255 μS	255 μS	255 μS
Vertical Serrations	No	Yes	Yes	Yes
H2 Inverter	None	None	Q7	A14-13
Auxiliary Video Pin	No	Yes	Yes	Yes
Optional 50 Hz Video	No	Yes	Yes	Yes
SYNC and COLOR REF on Slot 7	No	Yes	Yes	Yes
Solder Pads 7 and 8	No	No	Yes	Yes
HIRES Colors	4	6	6	6
Power-on Reset	No	Yes	Yes	Yes
Character Generator	2513	2513	2316B	2316B
Memory Jumper Blocks	Yes	Yes	No	No
Data Bus Transceiver	8T28	8T28	8T28	8304
SOFT 5	A2-8 & A2-11	A2-8 & A2-11	A2-8 & A2-11	R28

Single-Piece Keyboard

The single-piece keyboards used in early Apple II computers are identified by their physical arrangement. In the single-piece keyboard, the key switches and the ICs are all mounted on a common PC board. The functions performed on the single-piece keyboard are similar to those described in Chapter 7 for the two-piece keyboard. A different keyboard encoder IC is used, however, and there are other circuit differences. In this section, we provide a circuit description for the single-piece keyboard (Fig. C-21*).

(Note that *very early* single-piece keyboards use different reference designators than those shown in Fig. C-21*. These early keyboards use essentially the same parts and circuit, however. The early keyboards have a different PC layout that allows the ICs to be seen when the lid of the Apple is removed. This is how the early keyboards may be identified.)

Keyboard Encoder IC—The single-piece keyboard uses a type MM5740-AAE keyboard encoder IC, U5. The IC's 9 X outputs and 10 Y inputs are arranged in a matrix with the character keys at the cross points. Only 47 of the possible 90 cross points are occupied by keys. When a key is depressed, it connects the X line to the Y line.

Keyboard encoder U5 contains counters that count through an X sequence and a Y sequence as it scans the keyboard looking for a depressed key. These counters are driven from an external clock supplied to pin 3. An oscillator is formed by U4-6, U4-8, U4-11, C6, R6, and R7 that provides this clock. Its frequency (about 50 kHz) is not critical, but it must be between 10 kHz and 200 kHz for U5 to operate properly.

Fig. B-7 shows the clock and the sequence of pulses generated on U5's X outputs. When a key is depressed, one of the X outputs connects to a Y input. When this Y input is scanned, the key closure will be detected. Keyboard encoder U5 then looks at the SHIFT and CTRL key inputs and encodes the key into the proper ASCII character. The character is latched and appears inverted at U5's output (B1 through B9). The Apple II uses only the lower seven bits.

Gates U2 and U1-8 buffer the character and invert it to positive-true. The character then finds its way through the flat cable to location A7 on the mother board. Since the character was latched inside U5, it will remain at U5's output until another key is depressed.

Each time a key is closed, U5 outputs a pulse on its Data Strobe Output (pin 13). Pin 13 connects to pin 14 (Data Strobe Control), an arrangement that sets the pulse length at one clock period.

Sequence of Operation—Follow along on Fig. B-7 as we describe the timing. We will assume that the previous character was a "B." The ASCII LSB for "B" is 0, so U2-8 will be latched low from the previous character (point A). Now suppose you depress the "A" key. From the keyboard matrix we see that the "A" key connects X9 to Y4. Thus, as long as the "A" key is down, Y4 (point B) will be the same as X9 (point C). We will further assume that the Y scan is currently looking at Y4. This means the pulse on Y4 at point B will be detected by U5. Since the ASCII LSB for "A" is 1, U2-8 will go high (point D).

On the second clock falling pulse after detection (point E), U5 will generate a strobe pulse (point F). The pulse is one clock period long. The strobe is buffered by U1-6 and U1-3, then finds its way via the flat cable to location A7 on the mother board. There it sets flip-flop B10-9, point G. Chapter 6 covered the reading of the keyboard as far as circuitry on the mother board is concerned.

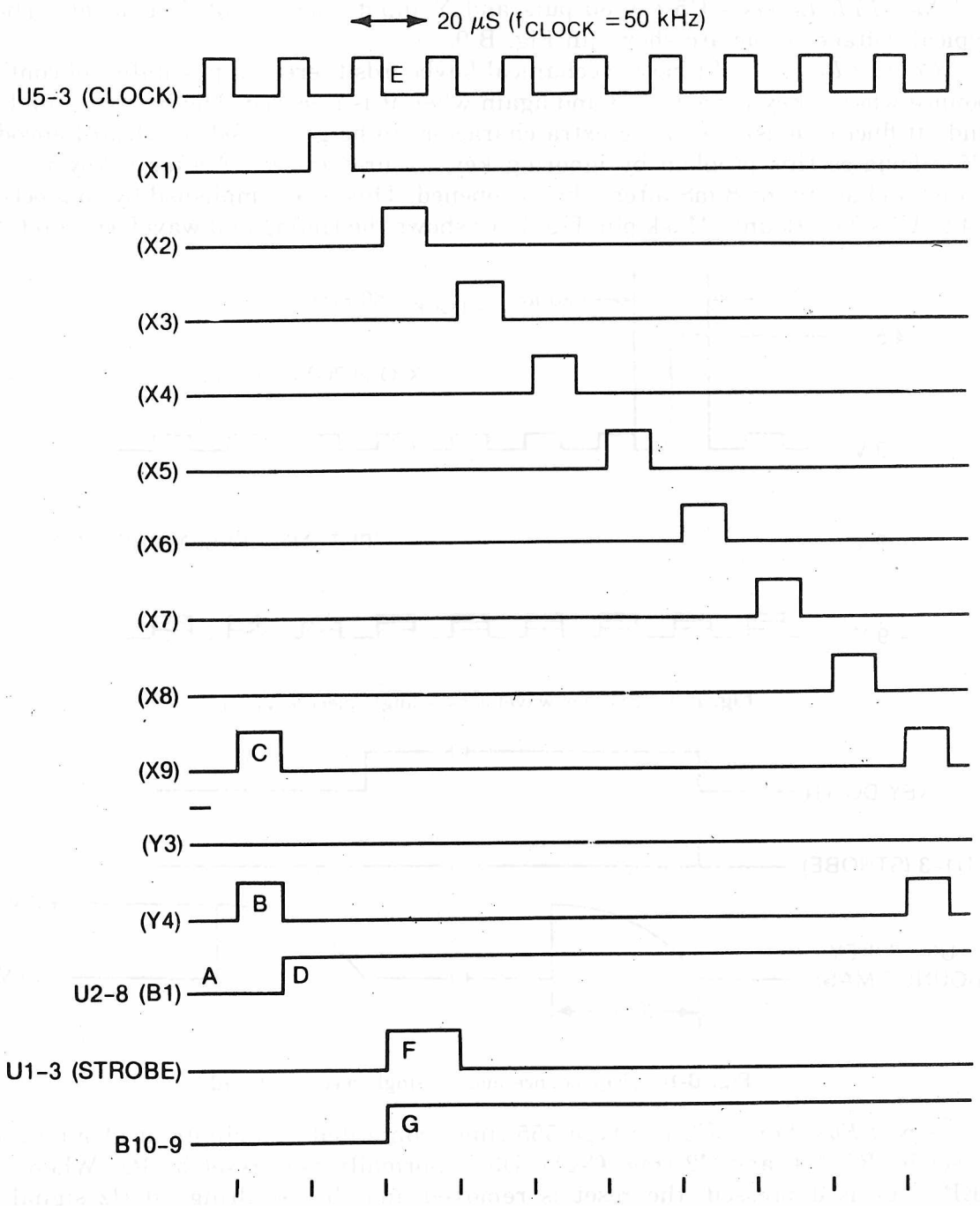


Fig. B-7. X scan—single-piece keyboard.

Fig. B-8* is another view of the signals shown expanded in Fig. B-7. In Fig. B-8* we can see one complete scan of the Y inputs. Again let's assume that "B" was the previous character so that B1 is low at point A. Sometime around point B we depress the "A" key. This connects Y4 to X9, and these two signals will be the same as long as the key is down. The key is not recognized until the Y scan reaches Y4 (point C). At this point. B1 goes high and there is a strobe pulse as was shown expanded in Fig. B-7.

The length of time it takes to detect a key closure is a function of where in the scan the key is depressed. The time ranges from 1 to 90 clock periods.

Non-TTL Levels—U5's X outputs and Y inputs are not at TTL levels. Their typical voltage levels are shown in Fig. B-9.

Contact Bounce—In most mechanical keyboards there is a possibility of contact bounce when a key is depressed and again when it is released. The bounce (rapid on and off fluctuations) can cause extra characters to be generated. Keyboard encoder U5 alleviates this problem by ignoring key closures for 8 mS after a key is first closed and again for 8 mS after a key is opened. This is accomplished by connecting C4 to U5's Key Bounce Mask pin. Fig. B-10 shows the timing and waveforms on C4.

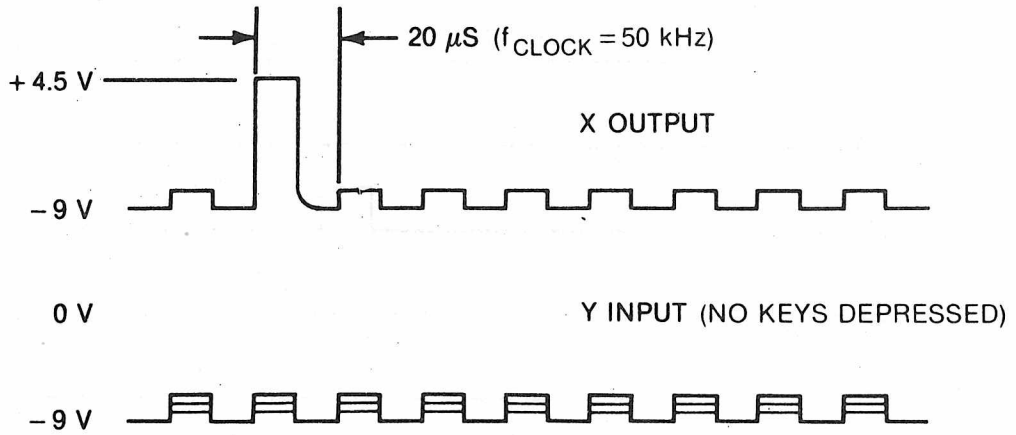


Fig. B-9. Encoder waveforms—single-piece keyboard.

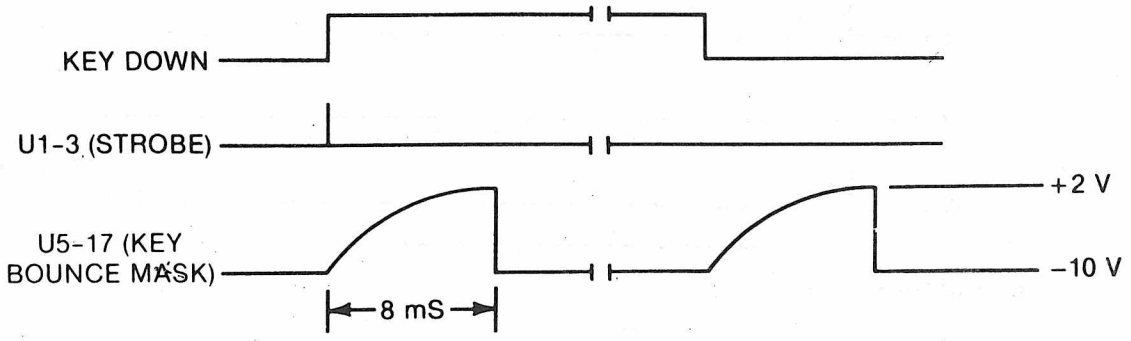


Fig. B-10. Key bounce mask—single-piece keyboard.

Repeat Function—U3 is a type 555 timer connected to oscillate at about 10 Hz as set by R3, R4, and C2 (Fig. C-21). U3 is normally held reset by R2. When the REPT key is depressed, the reset is removed and the resulting 10 Hz signal is applied to U5's Repeat input (pin 16). During each positive half cycle of the Repeat input, U5 will output a data strobe if a character key is also depressed.

Roll Over—The MM5740-AAE encoder IC is capable of *N-key roll over*. N-key roll over means that you can have any number of keys down simultaneously, and the encoder will correctly send the characters in the order keyed. This capability, however, can be used only if there is a diode at every cross point in the key matrix. The Apple II keyboard does not have these diodes. Without them, depressing three or more keys simultaneously may generate extraneous or *phantom* characters. As a result, the Apple II keyboard is described as having *2-key roll over*. No more than two keys may be depressed simultaneously if phantom characters are to be avoided.

The sneak paths that are possible in a matrix without diodes can be used in the

single-piece keyboard to generate characters not normally available from the keyboard as shown in Table B-2.

Table B-2. Generate Special Characters

Character	Hold down these keys simultaneously
_ (Underscore)	SHIFT, U, I, Y
[(Left Bracket)	SHIFT, U, I, J
\ (Backslash)	SHIFT, U, I, H

With the proper editing, you can insert these characters into programs directly from the keyboard.

Quad-Mode Operation—U5 is a quad-mode encoder. This means each key can generate up to four different characters as a function of the SHIFT and CTRL keys (see Table B-3).

Table B-3. CHARACTERS as a Function of Control and Shift

U5-19 (CTRL)	U5-21 (SHIFT)	Character Group
0	0	Unshift
0	1	Shift
1	0	Control
1	1	Shift-Control

The four characters for each key are shown in Fig. C-21*. Only the "P," "M," and "N" keys can generate four different characters. The rest generate one or two.

Initialization—C7 and R8 form a delay circuit that initializes U5 on power up by taking SHIFT high for about 5 mS. Diode CR1 provides a rapid discharge path for C7.